Nonelementary Reaction Kinetics. I

CENG252

Chemical Reaction Engineering

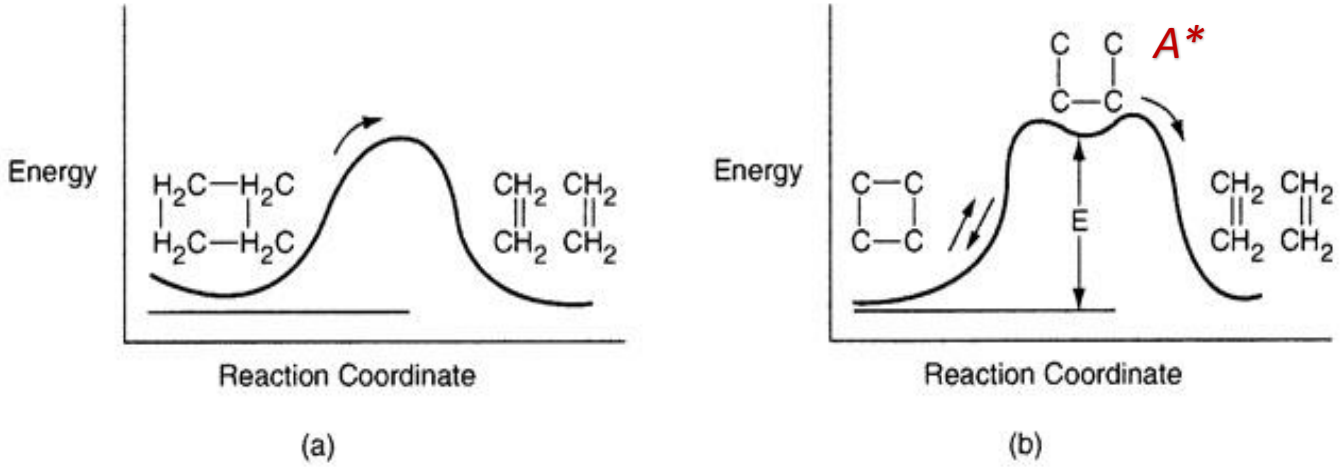
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Active Intermediate

- An ***active intermediate (A*)*** is an unstable molecule that is in a highly energetic and reactive state. It is short lived as it disappears virtually as fast as it is formed. It presents in very low concentrations.



Reaction coordinate. Courtesy *Science News, 156*, 247 (1999).

Pseudo Steady-State Hypothesis (PSSH)

- To apply the *pseudo steady-state hypothesis* assume that the net rate of formation of an active intermediate is zero.

- *The net rate of reaction of an active intermediate, A*, is zero.*

$$r_{A^*} = \sum r_{iA} = 0$$

Example 1:

The rate law for the reaction:

$$\mathrm{A} \rightarrow \mathrm{B} + \mathrm{C}$$

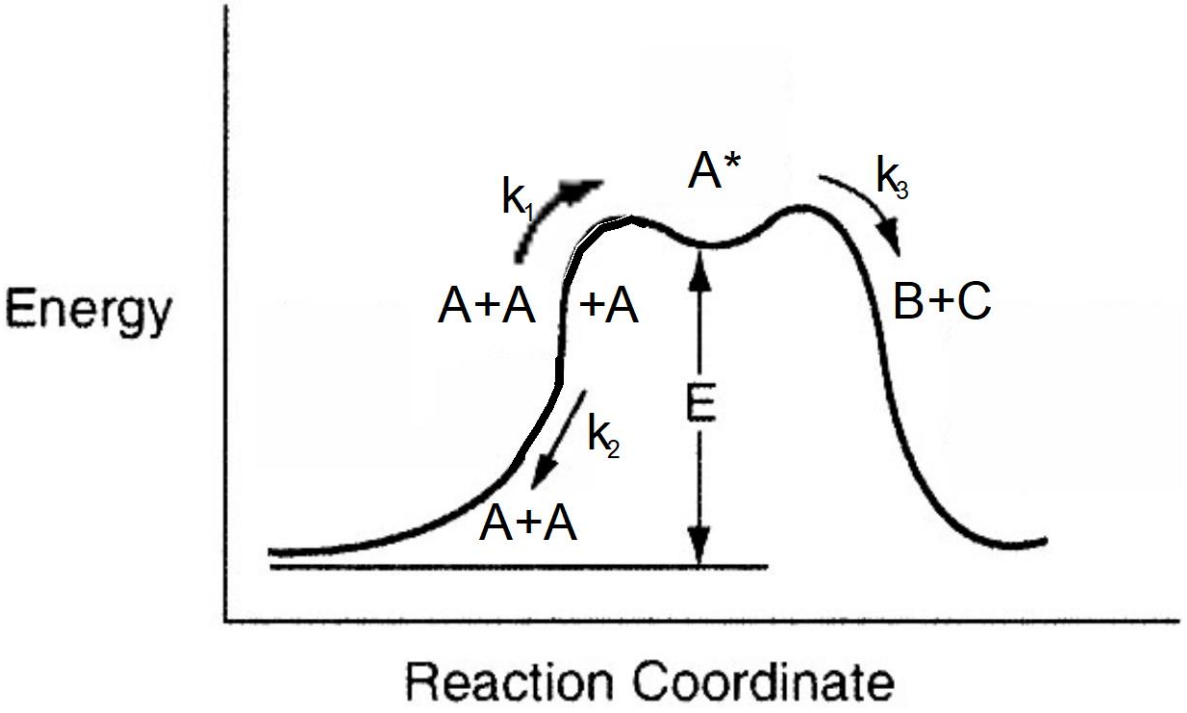
is found from experiment to be

$$-r_A = \frac{kC_A^2}{1 + k'C_A}$$

How did this rate law come about? Suggest a mechanism consistent with the rate law.

Example 1:

For reactions with active intermediates, the reaction coordinated now has trough in it and the active intermediate, A*, sits in this trough



Example 1 – Solution:

$$(1) \quad A + A \xrightarrow{k_1} A^* + A$$

$$(2) \quad A^* + A \xrightarrow{k_2} A + A$$

$$(3) \quad A^* \xrightarrow{k_3} B + C$$

Rate Laws:

$$(1)\quad A + A \xrightarrow{k_1} A^* + A$$
$$(2)\quad A^* + A \xrightarrow{k_2} A + A$$
$$(3)\quad A^* \xrightarrow{k_3} B + C$$

*k_3 is defined w.r.t. A**

Reaction (1) $$r_{1A^*} = k_1 C_A^2 \qquad (1)$$

Reaction (2) $$r_{2A^*} = -k_2 C_A C_{A^*} \qquad (2)$$

Reaction (3) $$r_{3A^*} = -k_3 C_{A^*} \qquad (3)$$

But C^*_A cannot be measured since it is so small

Relative Rates: $r_{1A} = -r_{1A^*}$, $r_{3B} = -r_{3A^*}$

Net Rates: Rate of Formation of Product:

$$r_B = r_{3B} = -r_{3A^*} = k_3 C_{A^*} \tag{4}$$

Pseudo Steady State Hypothesis: $r^*_A = 0$

$$r_{A^*} = \sum r_{A^*} = r_{1A^*} + r_{2A^*} + r_{3A^*} \tag{5}$$

$$= k_1 C_A^2 - k_2 C_A C_{A^*} - k_3 C_{A^*} = 0 \tag{6}$$

Solving for C_{A^*}

$$C_{A^*} = \frac{k_1 C_A^2}{k_3 + k_2 C_A} \tag{7}$$

Substituting for C_{A*} in Equation (4) the rate of formation of B is:

$$r_B = \frac{k_1 k_3 C_A^2}{k_3 + k_2 C_A} \tag{8}$$

Relative rates overall:

$$A \rightarrow B + C$$

$$\frac{r_A}{-1} = \frac{r_B}{1}$$

$$r_A = -r_B = -\frac{k_1 k_3 C_A^2}{k_3 + k_2 C_A} \tag{9}$$

$$r_A = -r_B = -\frac{k_1 k_3 C_A^2}{k_3 + k_2 C_A}$$

For high concentrations of A, we can neglect k_3 in the denominator with respect to $k_2 C_A$

$$k_2 C_A >> k_3$$

and the rate law becomes:

$$r_A = -\frac{k_1 k_3}{k_2} C_A = -k C_A \qquad (10)$$

(apparent first order)

$$r_A = -r_B = -\frac{k_1 k_3 C_A^2}{k_3 + k_2 C_A}$$

For low concentrations of A, we can neglect $k_2 C_A$ in the denominator with respect to k_3.

$$k_3 >> k_2 C_A$$

and the rate law becomes:

$$r_A = -\frac{k_3 k_1}{k_3} C_A^2 = -k_1 C_A^2 \qquad (11)$$

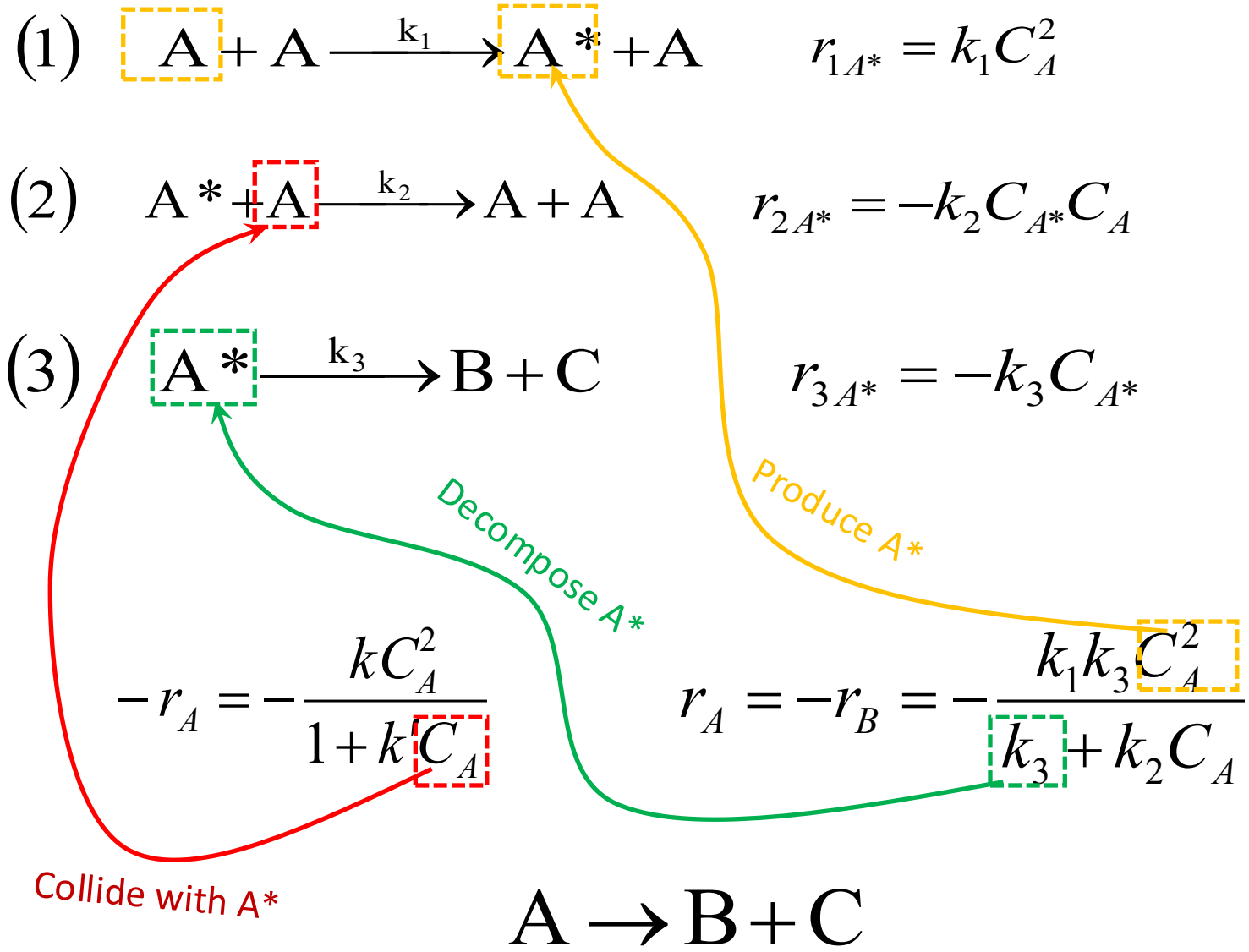
(apparent second order)

$$r_A = -r_B = -\frac{k_1 k_3 C_A^2}{k_3 + k_2 C_A}$$

Dividing by k_3 and letting $k' = k_2/k_3$ and $k = k_1$ we have the rate law we were asked to derive

$$\boxed{-r_A = \frac{k C_A^2}{1 + k' C_A}} \qquad (12)$$

Example 1 – Solution: $r_{A^*} = k_1 C_A^2 - k_2 C_A C_{A^*} - k_3 C_{A^*} = 0$



Active Intermediates

Why do so many reactions follow elementary rate laws?

What about $-r_A = kC_A$

$$-r_A = \frac{kC_A}{1 + k'C_I} \qquad (I = Inert)$$

(1) $A + I \xrightarrow{k_1} A^* + I$

(2) $A^* + I \xrightarrow{k_2} A + I$

(3) $A^* \xrightarrow{k_3} B + C$

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$$-r_A = \frac{k_1 k_3 C_A C_I}{k_3 + k_2 C_I}$$

Determining Reaction Mechanism

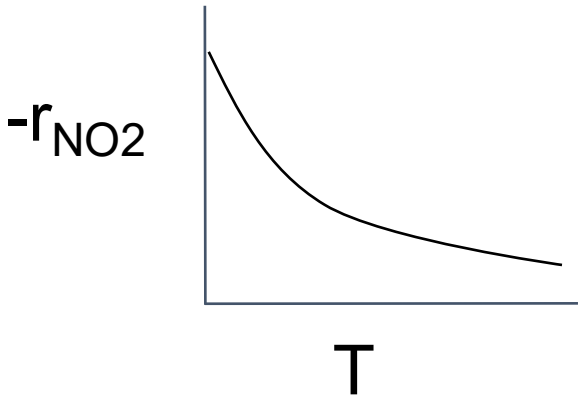
1. Assume an active intermediate(s)
2. Postulate a rate law (denominator contains species that collide with active intermediate)
3. Model each mechanism step as an elementary reaction
4. Write the rate laws for 3
5. Assume PSSH
6. Eliminate [A•] terms
7. Check for accuracy vs. experimental data

Rate Decreases with Temperature

- Consider the reaction: has an elementary rate law

$$2NO + O_2 \rightarrow 2NO_2 \qquad r_{NO_2} = k_{NO_2} C_{NO}^2 C_{O_2}$$

- However, rate *decreases* as temperature *increases, WHY?*



Rate Decreases with Temperature

- The reaction:

$$2NO + O_2 \rightarrow 2NO_2$$

- Occurs by the mechanism (elementary steps):

(1) $NO + O_2 \xrightarrow{k_1} NO_3^*$ $\qquad r_{1,NO} = -k_{1,NO} C_{NO} C_{O_2}$

(2) $NO_3^* \xrightarrow{k_2} NO + O_2$ $\qquad r_{2,NO} = k_{2,NO} C_{NO_3^*}$

(3) $NO_3^* + NO \xrightarrow{k_3} 2NO_2$ $\qquad r_{3,NO_2} = k_{3,NO_2} C_{NO_3^*} C_{NO}$

or $(r_{3,NO} = -k_{3,NO} C_{NO3} * C_{NO})$

Rate Decreases with Temperature

- By the PSSH:

$$r_{NO_3^*} = \sum_{i=1}^{3} r_{iNO_3^*} = 0$$

(1) $NO + O_2 \xrightarrow{k_1} NO_3^*$

$$r_{1,NO} = -k_{1,NO} C_{NO} C_{O_2} \qquad r_{1NO_3^*} = -r_{1,NO} = k_{1,NO} C_{NO} C_{O_2}$$

(2) $NO_3^* \xrightarrow{k_2} NO + O_2$

$$r_{2,NO} = k_{2,NO} C_{NO_3^*} \qquad r_{2NO_3^*} = -r_{2,NO} = -k_{2,NO} C_{NO_3^*}$$

(3) $NO_3^* + NO \xrightarrow{k_3} 2NO_2$

$$r_{3,NO_2} = k_{3,NO_2} C_{NO_3^*} C_{NO} \qquad r_{3NO_3^*} = \frac{-r_{3,NO_2}}{2} = -\frac{k_{3,NO_2}}{2} C_{NO_3^*} C_{NO}$$

Rate Decreases with Temperature

- Solving for the unknown active intermediate concentration:

$$r_{NO_3^*} = \sum_{i=1}^{3} r_{iNO_3^*} = 0$$

$$k_1 C_{NO} C_{O_2} - k_2 C_{NO_3^*} - \frac{k_3}{2} C_{NO_3^*} C_{NO} = 0$$

$$k_1 C_{NO} C_{O_2} - C_{NO_3^*}\left(k_2 + \frac{k_3}{2} C_{NO}\right) = 0$$

$$NO_3^* + NO \xrightarrow{k_3} 2NO_2$$

$$r_{3,NO_2} = k_{3,NO_2} C_{NO_3^*} C_{NO}$$

$$C_{NO_3^*} = \frac{k_1 C_{NO} C_{O_2}}{k_2 + \frac{k_3}{2} C_{NO}}$$

$$k_2 >> \frac{k_3}{2} C_{NO}$$

$$C_{NO_3^*} = \frac{k_1}{k_2} C_{NO} C_{O_2}$$

$$r_{NO_2} = k_{NO_2} C_{NO}^2 C_{O_2}$$

Nonelementary Kinetics

- What is the rate law for the following reaction?

$$H_2 + Br_2 \rightarrow 2HBr$$

$$r_{HBr} = \frac{k' C_{H_2} C_{Br_2}^{3/2}}{C_{Br_2} + k'' C_{HBr}}$$

H• and Br•

The end.

Recommended readings: **Chapter 7, Fogler 3rd Edition**